

Q1.

The waiting times for patients to see a doctor in a hospital can be modelled with a normal distribution with known variance of 10 minutes.

- (a) A random sample of 100 patients has a total waiting time of 3540 minutes.

Calculate a 98% confidence interval for the population mean of waiting times, giving values to four significant figures.

(4)

- (b) Dante conducts a hypothesis test with the sample from part (a) on the waiting times. Dante's hypotheses are

$$H_0 : \mu = 38$$

$$H_1 : \mu \neq 38$$

Dante uses a 2% level of significance.

Explain whether Dante accepts or rejects the null hypothesis.

(1)

(Total 5 marks)

Q2.

- (a) The length of one-metre galvanised-steel straps used in house building may be modelled by a normal distribution with a mean of 1005 mm and a standard deviation of 15 mm.

The straps are supplied to house builders in packs of 12, and the straps in a pack may be assumed to be a random sample.

Determine the probability that the **mean** length of straps in a pack is less than one metre.

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(4)

- (b) Tania, a purchasing officer for a nationwide house builder, measures the **thickness**, x millimetres, of each of a random sample of 24 galvanised-steel straps supplied by a manufacturer. She then calculates correctly that the value of $\bar{x}$ is 4.65 mm.

- (i) Assuming that the thickness, X mm, of such a strap may be modelled by the distribution $N(\mu, 0.15^2)$, construct a 99% confidence interval for μ .

(4)

- (ii) Hence comment on the manufacturer's specification that the mean thickness of such straps is greater than 4.5 mm.

(2)

(Total 10 marks)

Q3.

The weight of sand in a bag can be modelled by a normal random variable with unknown mean μ kilograms and known standard deviation 0.4 kilograms.

The sand in each of a random sample of 25 bags from a large batch is weighed. The mean weight of sand in these 25 bags is found to be 19.9 kg.

- (a) Construct a 98% confidence interval for the mean weight of sand in bags in the batch.

(4)

- (b) Hence comment on the claim that bags in the batch contain an average of 20 kg of sand.

(2)

(Total 6 marks)

Q4.

The weight, X kilograms, of sand in a bag can be modelled by a normal random variable with unknown mean μ and known standard deviation 0.4.

- (a) The sand in each of a random sample of 25 bags from a large batch is weighed. The **total** weight of sand in these 25 bags is found to be 497.5 kg.

- (i) Construct a 98% confidence interval for the mean weight of sand in bags in the batch.

(5)

- (ii) Hence comment on the claim that bags in the batch contain an average of 20 kg of sand.

(2)

- (b) The weight, Y kilograms, of cement in a bag can be modelled by a normal random variable with mean 25.25 and standard deviation 0.35.

A firm purchases 10 such bags. These bags may be considered to be a random sample.

- (i) Determine the probability that the **mean** weight of cement in the 10 bags is **less** than 25 kg.

(4)

- (ii) Calculate the probability that the weight of cement in **each** of the 10 bags is **more** than 25 kg.

(4)

(Total 15 marks)

Q5.

A random sample of 50 full-time university employees was selected as part of a higher education salary survey.

The annual salary in thousands of pounds, x , of each employee was recorded, with the following summarised results.

$$\bar{x} = 45.8 \quad \text{and} \quad s = 24.0$$

Also recorded was the fact that 6 of the 50 salaries exceeded £60 000.

- (a) Show why the annual salary, X , of a full-time university employee is unlikely to be normally distributed. Give numerical support for your answer. (2)
- (b) (i) Indicate why the mean annual salary, $\bar{X}$, of a random sample of 50 full-time university employees may be assumed to be normally distributed. (2)
- (ii) Hence construct a 99% confidence interval for the mean annual salary of full-time university employees. (4)
- (c) It is claimed that the annual salaries of full-time university employees have an average which exceeds £55 000 and that more than 25% of such salaries exceed £60 000.

Comment on **each** of these two claims.

(3)
(Total 11 marks)

Q6.

The volume of bleach in a 5-litre bottle may be modelled by a random variable with a standard deviation of 75 millilitres.

The volume, in litres, of bleach in each of a random sample of 36 such bottles was measured. The 36 measurements resulted in a mean volume of 5.05 litres and exactly 8 bottles contained less than 5 litres.

- (a) Construct a 98% confidence interval for the mean volume of bleach in a 5-litre bottle. (4)
- (b) It is claimed that the mean volume of bleach in a 5-litre bottle exceeds 5 litres and also that fewer than 10 per cent of such bottles contain less than 5 litres.

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Comment, with numerical justification, on **each** of these two claims.

(3)
(Total 7 marks)

Q7.

A factory produces bottles of brown sauce and bottles of tomato sauce.

- (a) The content, Y grams, of a bottle of brown sauce is normally distributed with mean μ_Y and variance 4.

A quality control inspection found that the mean content, y grams, of a random sample of 16 bottles of brown sauce was 450.

Construct a 95% confidence interval for μ_Y .

(3)

- (b) The content, X grams, of a bottle of tomato sauce is normally distributed with mean

μ_X and variance σ^2 .

A quality control inspection found that the content, x grams, of a random sample of 9 bottles of tomato sauce was summarised by

$$\sum x = 4950 \quad \text{and} \quad \sum (x - \bar{x})^2 = 334$$

- (i) Construct a 90% confidence interval for μ_X .

(5)

- (ii) Holly, the supervisor at the factory, claims that the mean content of a bottle of tomato sauce is 545 grams.

Comment, with a justification, on Holly's claim. State the level of significance on which your conclusion is based.

(3)

(Total 11 marks)

Q8.

- (a) The continuous random variable X has a rectangular distribution defined by the probability density function

$$f(x) = \begin{cases} 0.01\pi & u \leq x \leq 11u \\ 0 & \text{otherwise} \end{cases}$$

where u is a constant.

- (i) Show that $u = \frac{10}{\pi}$.

(2)

- (ii) Using the formulae for the mean and the variance of a rectangular distribution, find, in terms of π , values for $E(X)$ and $\text{Var}(X)$.

(2)

- (iii) Calculate **exact** values for the mean and the variance of the

circumferences of circles having diameters of length $\left(X + \frac{10}{\pi}\right)$

(4)

- (b) A machine produces circular discs which have an area of $Y \text{ cm}^2$. The distribution of Y has mean m and variance 25.

A random sample of 100 such discs is selected. The mean area of the discs in this sample is calculated to be 40.5 cm^2 .

Calculate a 95% confidence interval for μ .

(3)

(Total 11 marks)

Q9.

The waiting time at a hospital's A&E department may be modelled by a normal distribution with mean μ and standard deviation $\frac{\mu}{2}$.

The department's manager wishes a 95% confidence interval for μ to be constructed such that it has a width of at most 0.2μ .

Calculate, to the nearest 10, an estimate of the minimum sample size necessary in order to achieve the manager's wish.

(Total 5 marks)

Q10.

Cherie Glass is a salesperson for RDW Ltd, a replacement door and window company. Her supervisor, Wyn Doe, believes that Cherie spends too long discussing options with potential clients. With this in mind, he records the times, t minutes, that Cherie spends in discussions with 50 potential clients.

From these 50 times, Wyn finds that

$$\sum t = 3155 \quad \sum (t - \bar{t})^2 = 7180.5$$

- (a) Stating a necessary assumption about the 50 times, calculate unbiased estimates of the mean, μ , and the variance, σ^2 , for the time, in minutes, that Cherie spends with potential clients. Give your answers to one decimal place.

(3)

- (b) Hence construct a 99% confidence interval for μ .

(4)

- (c) Given that RDW Ltd expects its salespeople to spend, on average, no longer than 60 minutes with potential clients, comment on Wyn's belief.

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(2)

- (d) Given that $\mu = 62$, state the probability that a 99% confidence interval for μ will **not** contain 62.

(1)

(Total 10 marks)

Q11.

The times taken by new recruits to complete an assault course may be modelled by a normal distribution with a standard deviation of 8 minutes.

A group of 30 new recruits takes a total time of 1620 minutes to complete the course.

- (a) Calculate the mean time taken by these 30 new recruits.

(1)

- (b) Assuming that the 30 recruits may be considered to be a random sample, construct a 98% confidence interval for the mean time taken by new recruits to complete the course.

(4)

- (c) Construct an interval within which approximately 98% of the times taken by individual new recruits to complete the course will lie.

(2)

(Total 7 marks)

Q12.

The weight of gravel, in kilograms, collected by a power shovel may be modelled by a normal distribution with unknown mean, μ , and standard deviation 50.

The weights of a random sample of 20 collections have a mean of 1030 kg.

- (a) Construct a 98% confidence interval for μ , giving the limits to four significant figures.

(4)

- (b) Comment on a claim that the power shovel is, on average, collecting more than 1000 kg of gravel.

(2)

(Total 6 marks)

Q13.

- (a) The time taken, in minutes, by *Domesat* to install a domestic satellite system may be modelled by a normal distribution with unknown mean, μ , and standard deviation 15.

The times taken, in minutes, for a random sample of 10 installations are as follows.

47 39 25 51 47 36 63 41 78 43

Construct a 98% confidence interval for μ .

(5)

- (b) The time taken, Y minutes, by *Teleair* to erect a TV aerial and then connect it to a TV is known to have a mean of 108 and a standard deviation of 28.

Estimate the probability that the mean of a random sample of 40 observations of Y is more than 120.

(4)

(Total 9 marks)

Q14.

The height, in metres, of adult male African elephants may be assumed to be normally distributed with mean μ and standard deviation 0.20.

The heights of a sample of 12 such elephants were measured with the following results, in metres.

3.37 3.45 2.93 3.42 3.49 3.67 2.96 3.57 3.36 2.89 3.22 2.91

- (a) Stating a necessary assumption, construct a 98% confidence interval for μ .

(6)

- (b) The mean height of adult male **Asian** elephants is known to be 2.90 metres.

Using your confidence interval, state, with a reason, what can be concluded about the mean heights of adult males in these two types of elephant.

(2)

(Total 8 marks)

Q15.

- (a) The length, X centimetres, of adult male eels in a river may be assumed to be normally distributed with a mean of 38 and a standard deviation of 5.

Determine:

- (i) $P(X < 40)$;

(2)

- (ii) $P(30 < X < 40)$;

(3)

- (iii) the length exceeded by 75% of adult male eels in the river.

(4)

- (b) A sample of 40 adult female eels was taken at random from the river and the length of each eel was measured.

The mean and standard deviation of these lengths were found to be 107 cm and 19.1 cm respectively.

- (i) Construct a 98% confidence interval for the mean length of adult female eels in the river.

(4)

- (ii) Hence comment on a claim that, in this river, the average length of adult female eels is more than $2\frac{1}{2}$ times that of adult male eels.

(3)

(Total 16 marks)

Q16.

A machine is used to fill 5-litre plastic containers with vinegar. The volume, in litres, of vinegar in a container filled by the machine may be assumed to be normally distributed with mean μ and standard deviation 0.08.

A quality control inspector requires a 99% confidence interval for μ to be constructed such that it has a width of at most 0.05 litres.

Calculate, to the nearest 5, the sample size necessary in order to achieve the inspector's requirement.

(Total 6 marks)

Q17.

The weights of packets of sultanas may be assumed to be normally distributed with a standard deviation of 6 grams.

The weights of a random sample of 10 packets were as follows:

498 496 499 511 503 505 510 509 513 508

- (a) (i) Construct a 99% confidence interval for the mean weight of packets of sultanas, giving the limits to one decimal place.

(5)

- (ii) On each packet it states 'Contents 500 grams'.

Comment on this statement using **both** the given sample **and** your confidence interval.

(3)

- (b) Given that the mean weight of all packets of sultanas is 500 grams, state the probability that a 99% confidence interval for the mean, calculated from a random sample of packets, will **not** contain 500 grams.

(1)

(Total 9 marks)

Q18.

- (a) An analysis of a random sample of 40 customer visits to Lucy, a car saleswoman, results in a mean duration of 72 minutes and a standard deviation of 32 minutes.

Construct a 99% confidence interval for the mean duration of a customer visit to Lucy.

(4)

- (b) The time, Y minutes, that Ken, a car salesman, spends with a customer at a single visit has a mean of 53 and a standard deviation of 42.

- (i) Explain why a normal distribution is **unlikely** to provide an adequate model for Y .

(2)

- (ii) State why the distribution of $\bar{Y}$, the mean of a random sample of 60 single visits, is approximately normal.

(1)

- (iii) Hence determine the probability that $\bar{Y}$ is less than 60.

(4)

(Total 11 marks)

Q19.

The volume, in millilitres, of lemonade in mini-cans may be assumed to be normally distributed with a standard deviation of 3.5.

The volumes, in millilitres, of lemonade in a random sample of 12 mini-cans were as

follows.

155	148	156	149	147	156
157	156	150	154	148	154

- (a) Construct a 98% confidence interval for the mean volume of lemonade in a mini-can, giving the limits to one decimal place. (5)
- (b) On each mini-can is printed “150 ml”. Comment on this, using the given sample and your confidence interval in part (a). (3)
- (Total 8 marks)**

Q20.

A Trading Standards Officer suspects that the contents of a coal merchant’s 50 kg bags of coal are, on average, underweight.

He weighs the contents of a random sample of 10 bags of coal filled by the merchant. From these weights he calculates a mean of 49.2 kg.

- (a) (i) Assuming that the weights of the contents of bags of coal are approximately normally distributed with mean μ kg and standard deviation 1.6 kg, construct a 95% confidence interval for μ . (4)
- (ii) Hence comment on the Officer’s suspicion. (2)
- (b) The Officer decides that he requires the width of a 95% confidence interval for μ to be at most 0.75 kg.
- Find the minimum sample size to achieve this requirement. (4)

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Q21.

- (a) The volume, X millilitres, of olive oil in one-litre bottles may be assumed to be a normally distributed random variable with mean μ_X and standard deviation 3.
- (i) Assuming that $\mu_X = 1005$, determine the probability that the volume of olive oil in a randomly selected bottle is less than 1010 ml. (3)
- (ii) Find, to the nearest integer, the value of μ_X so that at most 1% of bottles contain less than 1 litre of olive oil. (4)
- (b) The volume, Y millilitres, of sunflower oil in one-litre bottles may be assumed to be a normally distributed random variable with mean μ_Y and standard deviation 3.

The volume, y millilitres, of sunflower oil in each of a random sample of 16 bottles

was measured, with the result that

$$\sum y = 16136.$$

Construct a 95% confidence interval for μ_Y , giving the limits to the nearest integer.

(5)

(Total 12 marks)

Q22.

The weight, in grams, of Italian grated cheese in cartons may be assumed to be normally distributed with mean μ and standard deviation 1.6.

From a random sample of 64 such cartons, the mean weight of grated cheese per carton is found to be 80.8 grams.

Construct a 95% confidence interval for μ .

(Total 4 marks)

Q23.

Nuts are dispensed into tins by a machine that can be set to deliver any specified mean weight. For any setting on the machine, the weights of nuts per tin are normally distributed with a standard deviation of 3.0 grams.

Twelve tins of nuts are selected at random from a batch filled by the machine at a particular setting. The weights, in grams, are recorded as follows.

413.1	412.6	410.4	410.6	415.2	414.6
416.6	411.2	416.9	412.6	413.4	408.8

- (a) Calculate, to one decimal place, the limits of a 99% confidence interval for the mean weight of nuts dispensed at this particular setting.

(5)

- (b) The setting on the machine is marked as “410 grams”. State, with a reason, whether this marking is likely to be correct.

(2)

(Total 7 marks)

Q24.

Pencils produced on a certain machine have lengths, in millimetres, which are normally distributed with a mean of μ and a standard deviation of 3.

A random sample of 16 pencils was taken and the length, x millimetres, measured for each pencil, giving

$$\sum x = 2848.$$

Construct a 99% confidence interval for μ .

(Total 5 marks)

Q25.

A hospital kitchen buys oranges in large batches. In each batch, the vitamin C content of the flesh of the oranges, in milligrams per 10 grams, may be modelled by a normal distribution with standard deviation 0.11.

The vitamin C contents of the flesh of a random sample of 8 oranges from a particular batch were measured, with the following results:

1.32 1.11 1.39 1.22 1.25 1.57 1.42 1.36

- (a) Calculate a 90% confidence interval for the mean vitamin C content of the flesh of oranges in this batch. (5)
 - (b) A nutritionist, who draws up diets for individual patients, assumes that the flesh of oranges has a vitamin C content of at least 1.20. Comment on this assumption as it relates to this batch of oranges. (2)
 - (c) The nutritionist requests that a 95% confidence interval of width not more than 0.03 should be calculated for the mean vitamin C content of the flesh of oranges in each batch. Calculate how large a random sample would have to be taken from each batch in order to achieve this. (5)
- (Total 12 marks)**

Q26.

A company manufactures components for the motor industry. The components are designed to have a length of 135.0 mm. Each day a technician takes a random sample of components and calculates a 95% confidence interval for the mean length.

- (a) On a particular Monday the technician takes a sample of nine components. Their lengths, in millimetres, are as follows.
 135.1 135.7 134.9 135.2 136.3 135.7 135.9 136.0 135.6
 Assuming the lengths of components produced on this Monday may be modelled by a normal distribution with standard deviation 0.42, calculate a 95% confidence interval for the mean length. Give your answer to an appropriate degree of accuracy. (6)
- (b) On a particular Friday a random sample of 60 components had a mean length of 135.8 mm and a standard deviation of 3.9 mm. Calculate a 95% confidence interval for the mean length of components produced on this Friday. (2)
- (c) The company overhauls all machines when the confidence interval calculated does not contain 135.0 mm. State whether increasing the sample size, n , is likely to increase, decrease or leave unaltered:
 - (i) the width of the confidence interval; (1)
 - (ii) the frequency of machine overhauls, explaining your answer. (2)

- (d) Using your results in parts (a) and (b) as an example, or otherwise, explain why it is possible that production on a day when the confidence interval does not contain 135.0 mm will sometimes be more satisfactory than production on a day when the confidence interval does contain 135.0 mm.

(2)

(Total 13 marks)

Q27.

A health food co-operative imports a large quantity of dates and packs them into plastic bags labelled 500 grams. Georgina, a Consumer Protection Officer, checked a random sample of 95 bags and found the contents had a mean weight of 498.6 grams, and a standard deviation of 9.3 grams.

- (a) Assuming that weights follow a normal distribution, calculate, for the mean weight of contents of all the bags:

(i) a 95% confidence interval;

(4)

(ii) an 80% confidence interval.

(2)

- (b) The health food co-operative also imports raisins. Georgina intends to take a random sample of 500 gram packets of raisins, weigh the contents and use the results to calculate an 80% and a 95% confidence interval for the mean weight, μ , of the contents of all the co-operative's packets of raisins.

(i) Find the probability that:

(A) the 80% confidence interval contains μ ;

(1)

(B) the 95% confidence interval contains μ but the 80% confidence interval does not.

(2)

- (ii) Instead of calculating both confidence intervals from the same sample, Georgina now decides to calculate the 95% confidence interval from one sample and the 80% confidence interval from a second independent random sample. Find the probability that the 95% confidence interval contains μ but the 80% confidence interval does not.

(2)

(Total 11 marks)

Q28.

Applicants to join a police force are tested for physical fitness. Based on their performance, a physical fitness score is calculated for each applicant. Assume that the distribution of scores is normal.

- (a) The scores for a random sample of ten applicants were

55 23 44 69 22 45 54 72 34 66

Experience suggests that the standard deviation of scores is 14.8.

Calculate a 99% confidence interval for the mean score of all applicants.

(5)

- (b) The scores of a further random sample of 110 applicants had a mean of 49.5 and a standard deviation of 16.5.

Use the data from this second sample to calculate:

- (i) a 95% confidence interval for the mean score of all applicants;
- (ii) an interval within which the score of approximately 95% of applicants will lie.
- (c) By interpreting your results in parts (b)(i) and (b)(ii), comment on the ability of the applicants to achieve a score of 25.
- (d) Give **two** reasons why a confidence interval based on a sample of size 110 would be preferable to one based on a sample of size 10.
- (e) It is suggested that a much better estimate of the mean physical fitness of all recruits could be made by combining the two samples before calculating a confidence interval. Comment on this suggestion.

(3)

(2)

(3)

(2)

(3)

(Total 18 marks)

Q29.

The weights of the contents of packets of table salt are known to be normally distributed with standard deviation 6 g. The mean weight of the contents of a random sample of 4 packets was found to be 748 g. Calculate a 99% confidence interval for the mean weight of the contents of all packets.

(Total 4 marks)

Q30.

A firm is considering providing an unlimited supply of free bottled water for employees to drink during working hours. To estimate how much bottled water is likely to be consumed, a pilot study is undertaken. On a particular day-shift, ten employees are provided with unlimited bottled water. The amount each one consumes is monitored. The amounts, in ml, consumed by these ten employees are as follows:

110 0 640 790 1120 0 0 2010 830 770

- (a) Assuming the data may be regarded as a random sample from a normal distribution, calculate a 95% confidence interval for the mean amount consumed on a day-shift.
- (b) (i) Give a reason, based on the data collected, why the normal distribution may not provide a suitable model for the amount of free bottled water which would be consumed by employees of the firm.

(8)

- (ii) A normal distribution may provide an adequate model but cannot provide an exact model for the amount of bottled water consumed. Explain this statement giving a reason which does not depend on the data collected. (3)
- (c) Following the pilot study the firm offers free bottled water to all the 135 employees who work on the night-shift. The amounts they consume on the first night have a mean of 960 ml with a standard deviation of 240 ml.
- (i) Assuming that these data may be regarded as a random sample, calculate a 90% confidence interval for the mean amount consumed on a night-shift. (4)
- (ii) Give **one** reason why it may be unrealistic to regard the data as a random sample of the amounts that would be consumed by all employees if the scheme was introduced on all shifts on a permanent basis. (1)
- (Total 16 marks)

Q31.

A manufacturer makes batteries for use in bicycle lights. The working lives, in hours, of the batteries are known to be normally distributed with a standard deviation of 1.8. A random sample of batteries was tested and their working lives were as follows:

48.2 49.6 47.1 50.0 46.8 47.2 47.9

- (a) Calculate a 95% confidence interval for the mean working life of the batteries. (6)
- (b) State the width of the confidence interval you have calculated. (1)
- (c) What percentage would be associated with a confidence interval of width 2 hours calculated from the given data? (5)
- (d) A further random sample of size seven is to be taken and used to calculate a confidence interval of width 2 hours. State the probability that this confidence interval will **not** contain the mean working life. (1)

(Total 13 marks)

Q32.

A manufacturer makes batteries for use in bicycle lights. The working lives, in hours, of the batteries are known to be normally distributed with a standard deviation of 1.8. A random sample of batteries was tested and their working lives were as follows:

48.2 49.6 47.1 50.0 46.8 47.2 47.9

- (a) Calculate a 95% confidence interval for the mean working life of the batteries. (6)
- (b) State the width of the confidence interval you have calculated. (1)

- (c) What percentage would be associated with a confidence interval of width 2 hours calculated from the given data?
- (d) A further random sample of size seven is to be taken and used to calculate a confidence interval of width 2 hours. State the probability that this confidence interval will **not** contain the mean working life.

(5)

(1)

(Total 13 marks)



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